Who Changes in College? Selection, and the Political Returns to Higher Education

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Abstract

Does higher education cause political change, or do graduates' choices reflect who was always going to enroll? This paper reframes the question as one of political returns, distinguishing engagement from partisan direction and asking how returns are distributed rather than estimating an average effect across the contrasting party systems of Germany and the United Kingdom. Attainment shifts graduates toward socially liberal and left-libertarian parties and away from the radical right. Political party systems, not the education systems, govern how this disposition reaches the ballot. Proportional representation converts left-libertarian support to a Greens vote in Germany, while first-past-the-post converts it into a Labour vote. These returns are unevenly distributed, concentrating among the most likely to attend in Germany and the least likely in the UK. Field-of-study differences divide graduates into STEM support for the center-right and HSS support for left-libertarians, exclusively among cohorts graduating after the financial crisis.

Keywords: Higher education; socialisation; comparative politics; selection; heterogeneity; political engagement; party identification; left party support; field of study; difference-in-differences; Germany; United Kingdom

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Higher education shapes political behaviour along two dimensions: graduates' political engagement and partisan support. The literature has become defined by the asymmetry between the two. Causal designs accounting for pre-adult socialization find selection accounts for the majority of the educational cleavage in civic and political engagement outcomes: political interest, turnout, participation, and civic activity (29; 39; 31; 59; 64). Yet when it comes to political *direction*, that is, graduates' political attitudes, orientations, and partisan support, the effects remain robust, substantively large, and often traceable to the field the graduates studied. Field-of-study effects on direction are documented in both Germany (27) and Britain (46), with field-based decompositions extending the finding to immigration attitudes across European democracies (17). Authors disagree about whether field effects operate on selection, content, or context (49; 28). The disagreement itself runs partly along national lines. This complicates the traditional literature on educational cleavages (26; 59): once selection is accounted for cleavage in political participation between graduates and non-graduates is small or absent. Political impacts on an individual seem to lie more among the graduates themselves in which side they are on and what they studied to find themselves there.

This asymmetry calls for a change in what I refer to as the *political returns* of higher education. The dominant frame, descending from Putnam (40) and developed in the resource models of Verba et al. (58); Brady et al. (6), treats higher education as an investment in civic skills that improve one's capacity to articulate, participate and engage politically. *Political returns* brings into view what civic returns leaves invisible: political interest as attitudinal disposition rather than behavioural input (51; 41; 64), party identification as attachment to a political community, and crucially, which party graduates support. Education seems to sort graduates towards supporting green and left-libertarians (1) with field of study explaining substantial variation in their immigration attitudes underlying these partisan choices (17). This points to a double sorting: politically engaged citizens sort into higher education, and higher education sorts them into different political camps depending on what they studied.

Changes among graduates by field face the same endogeneity concerns that attainment

does, after all, intuitively, a field is chosen in the same breath as choosing to attend university[2]. In causal research on attainment effects both liberal attitudes (47) and engagement largely dissolve to selection. Engagement differences appear already in adolescence prior to university attendance in Germany, the Netherlands and the United Kingdom (64; 54; 48). However, field of study variation is detectable under fixed effects (46; 28). Yet this recent work has not directly compare the contributions of attainment and field of study to the same variables across different educational contexts. This paper asks the question, between completing a university degree and studying the humanities or social sciences (or cultural and communicative content (55)), which contributes more to improving political engagement and partisan support? Any university credential or only certain fields of study?

If education is a proxy for prior political dispositions, the proxy works on the socio-cultural dimension and not on the economic dimension (47; 48). The educational cleavage has since the expansion of higher education been increasingly associated with new left-wing parties or issue-owning parties such as the Greens in Germany (1), rather than the old social democratic parties. These established associations are instructive because if there is a causal impact of tertiary education, this literature would expect the graduates to shift towards newer, issue-owning parties such as the green left-wing rather than redistributive social democratic parties. This leads to the prediction that, tertiary attainment leads to support for green left wing parties rather than support for left-wing parties generally.

This paper examines two theoretically distinct but inseparable educational choices as treatments: choosing to get a degree, and which field (operationalized as communicative and cultural fields versus economic and technical fields (CECT) for a detailed review, see Van de Werfhorst and Kraaykamp (55); Hooghe et al. (27)). Each one involves endogenous choice, moreover, the choices are predicted by mostly similar socio-demographic characteristics, but they operate on political outcomes through different mechanisms. This paper approaches these two choices using two datasets that complement each other well: The German SOEP (19) and the United Kingdom's

[2]I understand that this doesn't always apply to all national contexts, where you can decide your major a year after enrolling in university, however courses are also selected.

UKHLS/BHPS unified dataset (53). In the SOEP, field of study is observed as the actual completed field, capturing the effect of curricular exposure. In the UKHLS field of study is only recorded as a preference stated at age sixteen, this variable is unable to capture field effects, unless access to the National Pupil Database is obtained (16). The two countries also differ in their educational and electoral regimes. While Germany has early tracking and the UK has comprehensive mandatory education, both require academic secondary school qualifications: Abitur and A-levels. These two different country contexts provide fertile ground for investigating how the mechanism of tertiary and field effects can vary across political contexts.

Existing Research

The causal effect of higher education and fields of study on political behaviour and orientations remains intensely debated due to mixed empirical findings and the endogeneity issues that comes with estimating causal effects of educational choice (34; 49; 59). Identification strategies that seek to exploit "as if random" variation, such as regression discontinuity designs based on compulsory schooling laws (13), have yielded mixed results; while some detect small causal impacts of fields of study on political orientation (28), attitudes (18; 49; 46), or behaviour (2). The contested nature of these effects is well illustrated by the exchange between Kam and Palmer (29), who used a matching technique to argue that political participation was a spurious product of selection, and Henderson and Chatfield (23), who demonstrated that this result was driven by poor covariate balance and a handful of outlier matches. The underlying hurdle is the same in each case: attainment, attendance, and choice of field of study are all endogenous, and all can act as proxies for latent variables such as gender (14; 57), pre-adult socialization, and family background (39; 34; 47; 45).

Bourdieu (5) argues that higher education functions as a legitimising force of existing class hierarchies by sorting individuals into the social positions they are presupposed to occupy. Evidence of this thesis is found in the sorting model of education, or education-as-proxy argument (39; 59), wherein the only change after controlling for pre-adult factors reflects the existing social

hierarchy, a consequence of social sorting rather than an individual mechanism of socialization. According to Campbell (12) and Persson (38), the observed correlation between political engagement and tertiary attainment reflects the underlying characteristics of graduates more than it reflects a genuine causal impact. This is a compelling argument, but it is not a sufficient reason to conclude that socialization plays no role. Rather, it suggests that I cannot properly evaluate socialization without first modelling the heterogeneities of selection into higher education and how these vary within higher education across the fields of study people chose — treating these choices as integral to the causal process itself. Given the frequent finding that most liberalization processes are products of self-selection (47), it is worth investigating whether this self-selection is uniformly distributed across the likelihood of graduating higher education and across fields of study.

This brings us to a core theoretical problem. If I want to understand socialization, what higher education actually does to a person, I must first ask who is being socialized. The question of effect is inseparable from the question of selection, and selection into tertiary education is neither random nor uniform. The heterogeneity of who enters university, and under what conditions, shapes the distribution of any socialization effect I might hope to observe. Yet it remains an open empirical question whether selection patterns are themselves consistent with the socialization hypothesis (negative selection, whereby those least likely to graduate change the most) or the selection-as-proxy hypothesis (positive selection, those more likely to graduate change the most upon doing so because attainment amplifies pre-existing views or trends in engagement), or some combination of the two. Difference-in-difference models are limited in their capacity to investigate such patterns. This heterogeneity requires mapping a likelihood of graduation, and then using those stratifications to discover whether effects are uniform or concentrated among "particular types" of people as Simon (48) indicated.

As a recent review points out (60) the endogenous decisions in a causal process are integral to understanding the outcome, and requires investigation. The heterogeneous treatment effects of education on political behavior may stem not only from the educational experience itself but also from the endogenous selection into both tertiary education *and* field of study, both choices

significantly impacted by prior characteristics. The recent focus of political scientists engaging with the black box of endogeneity that is *causal effects of higher education* (47; 46) or *causal effects between fields of study* (34; 27) often leads to a pre-occupation with isolating, accounting for, abstracting from or establishing the relative contribution of self-selection. Selection in this context is understood in terms of the latent effect of individual background characteristics that can predict both political behaviour and these two educational choices. Yet, surprisingly, few have focused on variations in these political changes in terms of their background characteristics. In other words, I do not know what this selection is driven by; it is the estimated remainder, which I attribute to individual priors. Studying, for instance, the political change of randomly assigning students to study humanities is going to produce a different result than if they themselves had chosen to participate in such a class. Thus, I propose to focus on the student's decisions as part of the causal process, and using a prediction of their graduation to understand who is likely to be causally impacted by higher education, and who is likely to be impacted by their choice of discipline.

Tertiary Attainment and the Education–Politics Link

Brand (7) demonstrates that university education has non-uniform effects on the civic improvement I can expect from college completion. Specifically, improvements in civic engagement resulting from college completion decrease as the predicted likelihood of completing college increases. This pattern is consistent with what Brand and Xie (8) refers to as negative selection, that is, where individuals who are least likely to complete college but nonetheless do are most likely to improve. In simple terms, people who were unlikely to attend and graduate from university, but did anyway, gain the most from doing so. The comparison group consists exclusively of high school completers, as dropouts are excluded from the sample. The crucial aspect here is establishing a rich set of pre-treatment characteristics in order to define low and high propensities for completing university. Low propensity, for instance, was defined by parents with limited formal education, lower family income during adolescence, a non-intact family structure, having four or more siblings,

being Black or Hispanic, lower scores on prior cognitive indicators, enrollment in non-college-preparatory secondary school tracks, and having peers with limited educational plans or parents who did not encourage college enrollment.

The conclusions of Brand (7) support the Verba et al. (58) resource explanation in terms of the two resources that also predict skill acquisition (recall the three resources theorized to improve civic engagement by Verba et al. 1995: time, money, and skills). Those with a high propensity for college attendance also have more of these three resources, as well as an inherited politically engaged network. Their adolescent environments cultivate political interests earlier and more strongly than those of their low-propensity counterparts. A first-generation working-class tertiary graduate gains all three resources and a network, while a wealthy graduate student from a family of doctors is unlikely to experience the same change. Using this example, I can see how the differential access to resources and networks prior to tertiary attainment informs both the *decision* (selection) and the impact (causal). This differential has a further implication for the present moment. The share of parents who have attained tertiary degrees among students enrolling in higher education has naturally also expanded; one would therefore expect that in terms of within-individual change, the acquisition of civic skills has become *more* important than when the resource model was first theorized (58), precisely because the inherited resources that once did much of the work are less unevenly distributed across the student population than before.

Tracked Secondary Education and Field of Study

Before reviewing the literature on tracked education, two distinct concepts that structure the analysis should be clarified. Secondary school *track* refers to the institutional pathway through which students are sorted during compulsory education — Gymnasium versus Realschule/Hauptschule in Germany, A-levels versus vocational qualifications in the UK. Track determines who is eligible for university entry and is therefore central to the propensity score for tertiary attainment. *Field of study* refers to what a student studies once admitted, rather than whether they are admitted at all. The distinction between track (which determines *whether* someone enters university) and field

(which determines *what* they study once there) is analytically central: the paper estimates separate propensity scores and treatment effects for each decision.

With the expansion of tertiary education, I also identify new potential sources of heterogeneity, such as the theory that specific competencies within fields of study can socialize particular sets of attitudes or behaviors (55). The CECT framework scores a curriculum on four content dimensions, and the treatment is a single scalar derived from them: the share of a field's curricular content that is communicative or cultural, dichotomized at the sample median. It is one treatment, not two; summing the communicative and cultural dimensions is an aggregation assumption rather than a second intervention, a hidden-versions caveat carried through to the research design. Recent results have demonstrated these field effects among German graduates who majored in fields with greater communicative or cultural competencies (27). However, while they identify socialization effects on support towards Green-Alternative-Libertarian parties, their effect sizes are substantively minuscule. These effect sizes could be due to the fact that, in their estimation of attainment as an event, they average a dynamic treatment effect that is concentrated in a particular period across enrollment cohorts and periods. Contrasting graduates with non-attainers within a track-field strata is a compound treatment. The credential, the university experience in a certain period, and exposure to certain curricular content arrive together, while the control's own vocational or non-attaining trajectory, reflecting a lack of access or a choice not to proceed, is consequential to their politics. That design does not attribute its estimate to curricular content as such. Isolating content means holding attainment and the university experience fixed, comparing graduates of one field with graduates of another, and the counterfactual of the same graduate having studied something else. Other papers focusing on the content rather than the attainment alone find that content carries the partisan effects. For example, Martin et al. (33), who estimates effects of secondary-school subjects on party preference, and Simon et al. (49), who locates the attitudinal shifts from exposure to the Humanities and Social Sciences rather than in enrollment or graduation alone.

Research on tracked education systems reveals outcome-dependent findings regarding political socialization by academically or vocationally oriented tracks. When examining immigration

attitudes among Swiss youth from age 13 to 30, Lancee and Sarrasin (30) found no indication of tracked socialization patterns. Comparative studies (61) find some evidence linking secondary school track effects to higher education: tracked systems reduce enrollment in higher education, which reduces the pool of civic skills and political knowledge and thus civic and political engagement. Even among adolescents, strong selection patterns in political interest appear early. Zhang et al. (64) examined students by their track in the German National Educational Panel Study, showing that students had different levels of political interest *prior* to enrollment in upper-secondary education, with no significant causal effect of academic tracking on political interest thereafter. This is consistent with new data from the Dutch Adolescent Panel on Democratic Values, where Van De Werfhorst et al. (54) find that although students increase their political knowledge and trust in institutions in more advanced tracks, track differences in civic and democratic attitudes are already present before secondary education begins.

While the British comprehensive education system is not tracked to the degree that the German system is, it is, however, differentiated through the A-levels, which build on the generalized schooling of GCSEs. Similar selection patterns are found in research on the British higher education system, although with consistently modest leftward causal effects on cultural attitudes (34; 48; 45): moreover, students with culturally liberal priors disproportionately enroll in humanities and social sciences (HSS), while those with right-leaning economic views often select into STEM or business fields (49; 46). This university effect is further moderated by the system's high degree of mobility, since the vast majority of full-time students live in either university towns or more cosmopolitan areas, a similar environment conducive to peer socialization, moving towards the population political mean of their new neighborhoods (50). Similarly to the German case, the long-term political divide between graduates and non-graduates in Britain is attributed to several factors: selection, socialization, curriculum, and the allocation into new social networks and high-skilled workplaces that, for example, reinforce a less Eurosceptic and more culturally liberal worldview (34).

Hypotheses

The paper sets forth four predictions. Firstly, if educational attainment reflects only pre-adult dispositions, the treatment effect of tertiary attainment should collapse if I condition on pre-treatment political outcomes. Secondly, supposing there is a causal leftward shift from university graduation, I should observe support concentrating among socially liberal green and left-libertarian parties rather than a generalized left-wing support (1; Hix). If the findings by Brand (7) apply, treatment effects will mainly help graduates from disadvantaged backgrounds. However, if we see that more advantaged students benefit instead, it suggests a mechanism of cultural amplification rather than substitution. Finally, if the curricular content of communicative and cultural fields drives the partisan effect, the contrast between high-CECT and low-CECT graduates produces a partisan effect distinguishable from the attainment effect alone.

Table 1: Hypotheses

	Hypothesis
H1	Tertiary attainment has a causal effect on political behaviour, independent of pre-adult selection.
H2	University graduation shifts partisan support toward green and left-libertarian parties, not toward the left as a whole.
H3	The political return to attainment varies systematically with the likelihood of attaining a degree.
H4	A field's communicative-cultural content shifts partisan behaviour independently of attainment.

Research Design and Data

Rather than abstracting from the endogeneity of educational choice, this paper treats the endogenous choice as a meaningful inference. In tracked educational systems like Germany, tertiary attainment is strongly predicted by family background, gender, and migrant status. In the United Kingdom such predictions are weaker. If education's political consequences vary by the likelihood of treatment, that variation tells us something about how the institutional structure of accessibility determines socialisation into politics. The research design therefore asks how returns to higher ed-

ucation vary across predicted access to it, and how these returns vary between different institutional configurations of the same selection process.

What the average treatment effect would be under random assignment is not a question this design seeks to answer, nor is it one that can plausibly extend to the general population. Education is a service restricted and predicted by several factors, and the design treats these factors as substantively interesting sources of variation rather than something the estimator has to be robust to. Causal effects of higher education on political participation tend to converge towards a null finding (39; 47; 59). This is partly because of how aggregate ATTs are defined: we cannot recover effects from people whose counterfactuals would be quite different. Someone unlikely to attend university and someone almost certain to attend it experience and choose the same thing, tertiary education. Lindgren et al. (31) demonstrate that improving access to higher education increases turnout for lower-SES individuals and has no effect on higher-SES individuals. Brand and Xie (8) and Brand (7) frame the same intuition for higher education. This design treats this source of heterogeneity as an estimand rather than a nuisance parameter.

Two treatments are at stake, and keeping them distinct is essential to what each estimate means. The first is tertiary attainment itself: a binary, absorbing transition into higher education, against the counterfactual of the same person had they not enrolled. The second arises only among those who do enrol, and concerns the content of their degree: whether a communicative and cultural field rather than a technical or economic one moves graduates toward the left-libertarians (socially liberal and economically centrist parties) against the counterfactual of the same graduate had they studied the other kind of subject. The distinction matters because the second treatment has no untreated baseline. A graduate who reads English is no less treated by higher education than one who reads engineering; the relevant comparison is between two kinds of socialisation, not between a treatment and its absence. Socialisation occurs at university regardless of the field, however, these technical and economic fields are not exposed to the same degree of cultural, communicative or civic competencies. The two are answered by different estimators on different samples and do not share an identifying contrast, and the design develops them in turn: first the attainment effect

and its heterogeneity, then the field-content contrast within graduates.

Data

The analysis draws on two nationally representative household panels: the German Socio-Economic Panel (SOEP, 1984–2023) and the combined British Household Panel Survey and UK Household Longitudinal Study (BHPS–UKHLS, 1991–2022). Germany and the United Kingdom represent qualitatively distinct higher-education regimes, and this contrast is central to the research design. Both panels include youth questionnaires that provide pre-treatment political and educational measurements during adolescence. The data are restricted to cohorts born in or after 1980 in order to focus on the post-Bologna 1998 reform and the subsequent expansion of higher education (See Table 2).

Table 2: Analytical samples by specification

Specification	n_{treated}	n_{control}	Comparison
A. Attainment, eligibility-restricted (primary)			
SOEP — Abitur/Fachabitur, enrolled vs not	9,494	4,298	Eligible-enrolled vs eligible-not-enrolled
UKHLS — A-level, enrolled vs not	12,531	6,282	Eligible-enrolled vs eligible-not-enrolled
A′. Attainment, full never-treated control (robustness)			
SOEP	20,019	42,913	HE enrollment/completion vs no HE
UKHLS	12,531	26,455	HE-equivalent qualification vs no HE
B. Tertiary attainment by field track (SOEP)			
Tertiary | High-CECT track	4,738	3,167	HE effect within communicative-cultural
Tertiary | Low-CECT track	4,332	18,511	HE effect within economic-technical
C. Field of study within tertiary completers (SOEP)			
CECT-DE: high vs low CECT	4,534	4,536	Communicative-cultural vs economic-technical
Strict HASS vs non-HASS	2,590	5,763	Humanities/arts/social sci vs rest
Broad Liberal Arts vs STEM	3,829	4,524	+ teaching/caring/law/public order
HSS vs STEM+Economics	3,155	4,536	HASS + teaching + law vs STEM+Econ
D. Further robustness specifications			
SOEP — Tertiary (voc. controls)	20,019	7,481	HE vs vocational-training completers
SOEP — Tertiary (age ≤ 31)	12,680	27,899	Age restriction
SOEP — Tertiary (born ≥ 1980)	16,492	34,735	Narrower-cohort robustness

Notes: Unique persons in each specification; outcome-specific cells in partisan-direction analyses are smaller because they condition on reporting party support. In the UKHLS, every degree-holder is A-level-eligible by construction, so the eligibility restriction in Panel A trims only the control group (26,455 to 6,282), not the treated. In Panel A the eligible-not-enrolled control mixes genuine non-appliers with applicants who were not admitted (see limitations).

Treatments

Tertiary attainment

The attainment treatment is a binary, absorbing, within-person indicator that switches to 1 from the year of first observed higher-education enrollment and remains so. The operationalisation is harmonised across SOEP and UKHLS. The control group is the never-treated population. In Germany this encompasses Abitur and Fachabitur holders who did not proceed to university, those who completed vocational training (Ausbildung) without subsequent tertiary enrollment, and those whose highest qualification is upper-secondary or below. In the United Kingdom the equivalent group comprises A-level holders who did not proceed to higher education and respondents at ISCED-4 or below. Any respondent ever observed enrolled in higher education is in the treated group. In the UKHLS, where the enrolment year is not recorded directly, holding a higher-education qualification also marks treatment from the wave it first appears.

The primary attainment comparison restricts the sample to the higher-education-eligible: Abitur or Fachabitur holders in the SOEP and A-level holders in the UKHLS (Table 2, Panel A). Within this restriction the comparison is eligible respondents who enrolled against eligible respondents who did not. The restriction matters because the broader never-treated group mixes respondents who never crossed the academic threshold to higher education with respondents who did. The eligibility-restricted counterfactual is the same person, already academically prepared, had they not enrolled, which is closer to isolating the effect of enrollment itself than the broader contrast between prepared and unprepared respondents.

Cultural-communicative content

The field treatment is observable only in the SOEP[3]. Field is classified by the CECT scheme (Communicative, Economic, Cultural, Technical) of Van de Werfhorst and Kraaykamp (55), dichotomized at the within-sample median of the CECT ratio as described by graduates of each de-

[3]Field of study is recorded in the SOEP after a degree is obtained; it is carried forward and backward within individuals to align with the absorbing-attainment indicator.

gree. The CECT ratio is the share of communicative and cultural skill content relative to the sum of all four skill dimensions, with skill weights taken from Van de Werfhost and De Graaf (56)'s graduate survey (formula in Appendix). Three content-based classifications are reported alongside the median split as robustness test following recent literature on field of study effects on political attitudes (17; 32; 46; 49). A strict HASS (humanities, arts, social sciences), a broader Liberal Arts grouping (HASS with teaching, caring professions, law, and public-order professions), and an HSS grouping (HASS with teaching and law, excluding caring and public-order professions).

The field contrast sets high-CECT against low-CECT, or HASS-content against STEM and economics under each content grouping (Table 2, Panel C). Because each graduate is measured against their own pre-enrollment baseline, differencing the high-CECT trajectory against the low-CECT trajectory nets out the common effect of attending university and leaves the differential change attributable to communicative-cultural rather than technical-economic content. Panel B of Table 2 reports the attainment-within-track contrast that follows Hooghe et al. (27); it is retained as a track-heterogeneity check on the attainment effect rather than as the field estimand.

Outcomes

I examine political engagement and behaviour as two dimensions of political returns to higher education and field of study, with covariates harmonised across the SOEP and UKHLS using the manual developed by Turek and Kalmijn (52).

The first dimension concerns two engagement items, political interest and party identification. *Political interest* is measured annually on a four-point scale in both countries (1 = not at all interested, 4 = very interested). *Party identification* is binary: separately from which party they identify with, the respondent indicates whether or not they have a party they support.

The second dimension relates to the direction of political behaviour. I use political party support variables in combination with retrospective voting record and party identification variables in the UKHLS and SOEP to distinguish supporting a party from voting for a party. This illustrates the slight discrepancy between waves in terms of which party panel respondents recall they voted

for and what party they supported in the year the election was called. Concordance between lagged support and recall vote choice ranges from 0.89 to 0.95. Throughout, the "Green/lib-left" party-support outcome refers to the broader green and left-libertarian grouping in Germany rather than to just the Greens: Bündnis 90/Die Grünen, Bürgerrechtsbewegungen, ÖDP, Piraten, and other green or left-libertarian small parties, matching the grouping of parties in Hooghe et al. (27); in the UK first past the post system the party capturing similarly socially liberal and economically centrist (left-libertarian) party are the Liberal Democrats.

To investigate party support in terms of respondents' realised political behaviour I constructed an annual binary composite for each party. In the SOEP the composite has three tiers: a recall vote item from the waves following three general elections in Germany (waves 2014, 2018, and 2022, `plh0333`: "Which party did you vote for in the 2013 / 2017 / 2021 Bundestagswahl?"); a party support item conditional on a yes to the engagement question about party identification at all other annual waves (`plh0011`: "Generally speaking, do you lean towards a particular party?"; followed by `plh0012`: "Which party do you support?"); and zero for respondents reporting no party identification. In the SOEP respondents were asked to indicate which party they voted for; in the UKHLS respondents were asked if they voted in the last general election (`vote9`: "Did you vote in the most recent general election?"). The UKHLS composite is constructed from a combination of party support (`vote1`: "Generally speaking, do you think of yourself as a supporter of any one political party?"; followed by `vote4`: "If there were a general election tomorrow, which political party do you think you would be most likely to support?") plus a commitment proxy where respondents have reported which party they support prior to a wave in which they participated in an election. Respondents in both datasets are therefore required to have voted in at least one general election.

Identification strategy

The design follows the standard approach of estimating aggregate treatment effects before investigating heterogeneity. Formal specifications and identifying assumptions for each estimator are in

Appendix .

The aggregate effect of attainment and its heterogeneity

The aggregate attainment effect on engagement and partisan direction is estimated using the Callaway and Sant'Anna (10) staggered difference-in-differences estimator with inverse-probability weighting (IPW), never-treated controls, and bootstrapped standard errors. Respondents are under treatment from the year they first enrol in higher education. Weights are constructed on predictors of higher education: gender, parental education, migrant background, and year of birth. Cohort-time treatment effects are aggregated to an overall ATT of attending university (10).

For the United Kingdom, the same outcomes are estimated using the Xu et al. (62) interactive fixed effects (IFE) estimator on the per-wave absorbing status indicator, with covariates gender, ethnicity, parental education, and year of birth. UKHLS records the first wave at which a qualification is reported rather than the year of enrollment and graduation. The fixed-effect counterfactual is heterogeneity-robust and does not depend on cohort decomposition. The two estimators have distinct identifying assumptions: parallel trends conditional on covariates and no anticipation of treatment effects in the cohort decomposition for Germany, and the imputation of counterfactual parallel trends for the United Kingdom. I test both using pre-trend tests and a placebo equivalence test on at least two periods prior to enrollment (42).

In addition to the aggregate analysis on party support, the composite measure (Outcomes section) is estimated using IFE on both samples to test for discrepancies between support and composite ATTs. This identifies whether graduates' attachment to a party translates into voting for that same party. The composite ATT is also compared to the cohort-specific ATTs from the staggered specification to see whether any support-vote discrepancy is uniform across treatment cohorts or concentrated in particular cohorts, providing a cohort-replacement view of how attachment maps onto vote choice.

Heterogeneity in the tertiary attainment effects are estimated in Germany, thanks to a useful instrument, using the marginal treatment effect framework of Zhou and Xie (66). The goal here

is to be able to describe how political returns to a degree vary by how likely a person was to obtain one. Brand (7) answered this in terms of civic returns by sorting people into propensities of attainment and then comparing effects between buckets of propensities. The problem as stated by Zhou (65) is that this requires the propensity score to capture everything that drives both attendance and the outcome; anything unmeasured, such as luck, drive, institutional barriers, or grit leaks into the strata and contaminates the comparison. The solution to this is accounting for unobserved resistance and predictions of attainment on demographics, family background, and the secondary school track. The interesting population lies at the margin, close in both propensity and resistance, and is the subpopulation that would respond to a change in the accessibility of higher education.

To estimate this marginal effect Zhou and Xie (66) uses a local instrument, for attainment the birth Bundesland is chosen, which shifts the probability of attendance through the regional supply of places and regional higher-education policy but is assumed not to affect adult party support except through whether the respondent attended. Comparing respondents pushed just over the attendance threshold by the region they were born in against otherwise-similar respondents just under it isolates the effect on the marginal attendee; sweeping the instrument across its range traces out the treatment effect as a function of resistance. Three population-average effects are reported: the effect on those who attended (the treated), the effect on the untreated had they attended, and the effect on the marginal subpopulation. A treatment effect that is larger for the untreated than the treated, equivalently an effect that rises with resistance, indicates negative selection on the gains from higher education: the people least likely to attend would have benefited most. This is the equalisation pattern of Lindgren et al. (31) and the negative-selection pattern of Brand and Xie (8). The reverse ordering indicates positive selection.

The UK data does not have an equivalent instrumental variable (the tuition-fee instrument failed the relevance test). The treatment effect by likelihood is approximated by three converging data-driven decompositions. DiD ATTs within propensity quintiles implement the Brand (7) model in modern DiD form. The IFE estimator is refit within the A-level subpopulation, the strongest predictor of HE attainment. Finally, IFE is estimated within enrollment cohorts, since early enrollees

tend to be high-propensity relative to later enrollees within a cohort. Convergence of the three decompositions provides non-parametric evidence of heterogeneity-by-likelihood, applying a similar logic to the marginal treatment effect framework (66).

A descriptive estimate complements the propensity heterogeneity. I estimate the within-ATT using augmented inverse-probability-weighted interaction estimators on each treatment-outcome-moderator separately. Moderators are gender, high parental education, migrant or ethnic background, and high parental occupational status. This identifies the mechanism described in Lindgren et al. (31) that electoral returns to tertiary attainment are largest among respondents whose family background makes them unlikely to attain a degree.

Field of study among graduates

Identification rests on conditional ignorability of field choice given the observable propensity score and pre-tertiary political attitudes, the selection-on-observables analogue of the instrument-identified attainment heterogeneity above.

The estimator is the same staggered difference-in-differences design used for attainment, with the treatment redefined as the high-CECT (or HASS) indicator and the comparison restricted to graduates. The cohort decomposition is substantively important here. The IFE design of Hooghe et al. (27) averages a dynamic ATT across all cohorts at each event-time, collapsing cross-cohort variation in the field effect into a single curve. The staggered DiD identifies $\mathrm{ATT}(g,t)$ at each cohort separately by year of enrollment and aggregates to an overall ATT and cohort-summary ATT using explicit, non-negative weights over the cohort-time grid (20). The decomposition is useful for understanding qualitative change in educational expansion: HASS graduates from the 2005 treatment cohort can have a different ATT than those from the 2018 cohort. Both estimators avoid the negative-weighting issues of two-way fixed effects (3; 43). The UK panel does not record completed field with sufficient coverage to support a parallel analysis.

Compound treatments and limitations

Three compound-treatment concerns warrant explicit acknowledgment. First, the field contrast bundles content with peer composition and post-graduation income trajectory. The within-graduate ATT identifies the joint effect of the bundle, not the marginal effect of curricular content alone.

Second, the attainment control group includes respondents whose higher-education eligibility differs from the treated. The eligibility-restricted specification (Panel A) addresses this by conditioning on the academic gate. Within the eligible never-enrolled group, however, the survey questions do not distinguish genuine non-appliers from respondents who applied and were not admitted, and the application records have too little coverage to support an individual-level decomposition. The compound-treatment effect of application and rejection is therefore not identified and remains bundled into the never-enrolled control. Pre-treatment political interest is conditioned on in the propensity weights, which absorbs any systematic difference in baseline engagement between non-appliers and rejected applicants, leaving the residual bias attributable to within-attitude-bin heterogeneity in compound-treatment exposure.

Third, the field contrast does not have a field-margin instrument. Birth Bundesland predicts overall higher-education availability but not the relative supply of HASS versus STEM and economics places, so a parallel local-IV identification on the field margin is not available with the present data. The within-graduate Panel C ATTs are therefore identified by conditional ignorability of field choice rather than by an exclusion restriction; they recover the contrast on the marginal subpopulation close to indifference between fields, and cannot be extrapolated to all graduates.

A final concern is missingness, of two kinds. I test panel attrition for differential dropout by treatment and find it ignorable conditional on the covariates, so the within-person estimates are not driven by purely by panel or treatment drop out. The lean party-support outcomes are a sharper concern: they are observed only for respondents who report a party, a set that tertiary attainment plausibly enlarges through its own engagement effect, so conditioning on it selects on a consequence of treatment. The composite vote-proxy outcomes sidestep this by coding non-identifiers as zero and are therefore estimated on the full sample; the agreement between the lean

and composite estimates bounds the selection impacts.

Results

Aggregate effects on engagement and party support

Beginning with the causal estimates in the aggregate, Table 3 reports the ATT of tertiary attainment on engagement and party support in both panels, estimated with the interactive fixed-effects counterfactual of Xu et al. (62). Getting a university degree raises engagement by a few percentage points in both countries, but the two panels do not move on the same outcome. In Germany the gain is on political interest and is precisely estimated; in the UK political interest does not move detectably, while party identification rises further than it does in Germany. The German pattern is consistent with the argument that political interest is socialised and stabilized much earlier in the life course than during higher education (64): pre-university political interest is largely determined by family background, demographics and parental social status (B.3, withheld), which leaves little for the degree itself to move. Where the education system has done less sorting, as in the UK, the entrant's baseline is less determined and the return appears on attachment to a party rather than on interest in politics.

On the partisan returns, British graduates move towards the Liberal Democrats rather than the historically social-democratic Labour party. In Germany the pooled estimates demonstrate no significant partisan movement on any party; the staggered decomposition (Table 4) recovers a shift toward the Greens, concentrated in particular enrollment cohorts (Figure 4). This supports our theoretical motivation for using a staggered-DiD estimator over the pooled event-study, the two moving time bounded targets of political parties as they define themselves dynamically, and how the university experience changes over a twenty year period have a significant impact on the estimation of tertiary effects. The political returns towards the Greens in Germany is a cohort phenomenon local to the period in which the entrants graduated. That the event-study estimator returns no partisan movement is not evidence that attainment leaves German partisanship untouched. A null can mark the absence of an effect or a mismatch between the window over which an effect is measured and the window over which it operates (36; 15). The staggered approach enables a

cohort decomposition, which can differentiate between the two and ultimately supports the latter.

The composite vote-proxy ATT (Table 5) anchors party leaning in actual vote choice and clarifies a discrepancy in the UK partisan results. For the Labour party, the composite vote-proxy ATT substantially exceeds the support-leaning ATT. Graduates' Labour support lags behind how they actually vote; this largely reflects strategic voting in a historically two-party, first-past-the-post system. In Germany, leaning and the composite vote-proxy align more closely.

Attainment also moves graduates *away* from the radical right, and here the comparative approach resolves an ambiguity. In Germany, tertiary attainment reduces overall AfD support, but this effect becomes statistically insignificant because Abitur graduates so rarely support the AfD that the null is indistinguishable from a floor effect. The comprehensive UK system allows for a clean comparison because far-right (UKIP/Reform) support has a much higher base rate, large enough for precise estimates. The UK reproduces the same German pattern with greater statistical power: tertiary attainment reduces far-right support, but the reduction is concentrated entirely among those who did not obtain their A-levels (Table 6). The UK null among A-level holders shows that what could appear as an inoculating effect of the university is actually a reflection of school-track selection common to tracked and comprehensive systems, consistent with UK-based evidence that political sorting precedes higher education (32; 49) rather than being produced by the university itself.

How tertiary effects vary by likelihood of attaining a degree

I identify the attainment effect on engagement and partisan support by likelihood of graduation, i.e. the political returns to higher education (in a similar vein as (8)). Two complementary identifications recover this heterogeneity: a local-IV implementation of the marginal treatment effect for Germany using birth Bundesland as instrument, and a three-way non-parametric decomposition for the UK in the absence of good instrument candidates in the data.

Table 7 reports the local-IV estimates for Germany following the Zhou and Xie (66) selection-on-unobservables criticism of the original heterogeneous treatment estimator (8). The effect on

Table 3: Aggregate effects of tertiary attainment on engagement and party support

	Engagement		*Party support*			
	Political interest	Has party	Centre-right	Centre-left	Left-libertarian	Other left
Germany — interactive fixed-effects counterfactual						
Tertiary (Univ)	$+0.048^{***}$	$+0.058^{***}$	$+0.001$	$+0.014$	$+0.010$	-0.001
	(0.010)	(0.008)	(0.011)	(0.011)	(0.011)	(0.008)
United Kingdom — interactive fixed-effects counterfactual						
Tertiary (Bachelor)	$+0.039$	$+0.066^{***}$	-0.030	-0.032	$+0.055^{**}$	$+0.001$
	(0.028)	(0.016)	(0.022)	(0.024)	(0.021)	(0.012)

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Bootstrap standard errors in parentheses.
Outcome columns: Centre-right = CDU/CSU (DE) / Conservative (UK); Centre-left = SPD (DE) / Labour (UK); Left-libertarian = Greens (DE) / Liberal Democrat (UK); Other left = Linke (DE) / Green Party (UK).
`fect` interactive fixed effects (Xu et al. 62) on the absorbing tertiary-status indicator, ages 14–31 (matrix completion for the German engagement outcomes). Covariates: gender, parental education, migration/ethnicity, parental ISEI (Germany).

Table 4: Germany, Callaway–Sant'Anna staggered DiD (cohort-decomposable robustness)

	Engagement		*Party support*			
	Political interest	Has party	CDU/CSU	SPD	Greens	Linke
Tertiary (Univ)	$+0.037^{*}$	$+0.046^{***}$	-0.020	$+0.023$	$+0.037^{**}$	-0.007
	(0.020)	(0.012)	(0.019)	(0.017)	(0.018)	(0.010)

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Bootstrap standard errors in parentheses.
`did::att_gt` (Callaway and Sant'Anna 11), IPW, never-treated control, universal base period. Decomposing by enrolment cohort recovers a Greens shift that the pooled estimate (Table 3) averages toward zero, concentrated in the 2005–09 cohort (Figure 4).

Table 5: Vote-proxy composite: interactive fixed-effects ATT, both panels

	Centre-right	Centre-left	Left-libertarian	Other left
Germany				
Tertiary (Univ)	$+0.002$	-0.015	$+0.036^{**}$	$+0.007$
	(0.014)	(0.014)	(0.016)	(0.008)
United Kingdom				
Tertiary (Bachelor)	$+0.003$	$+0.049^{***}$	$+0.013^{**}$	$+0.019^{***}$
	(0.011)	(0.016)	(0.006)	(0.005)

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Bootstrap standard errors in parentheses.
`fect` on the annual vote-proxy composite (recall vote at election waves, party leaning otherwise). Columns as in Table 3; Other left = Linke (DE) / Green Party (UK).

Table 6: Tertiary attainment and far-right support, by academic eligibility (both panels)

	Overall	Academically eligible	Not eligible
Germany (AfD)	-0.034^{***}	-0.008	—
	(0.005)	(0.005)	
United Kingdom (UKIP/Reform)	-0.003^{*}	-0.000	-0.022^{**}
	(0.001)	(0.001)	(0.007)

ATT of tertiary attainment on far-right support. Germany: CS-DiD (IPW, never-treated); UK: Xu et al. (62) fect. "Academically eligible" = Abitur/Fachabitur (DE) or A-level (UK) holders. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. The German non-eligible cell is not estimated separately; the overall-vs-eligible gap implies the reduction is carried by the non-eligible, confirmed directly in the UK. The reduction tracks the secondary-level eligibility gradient, not university attendance.

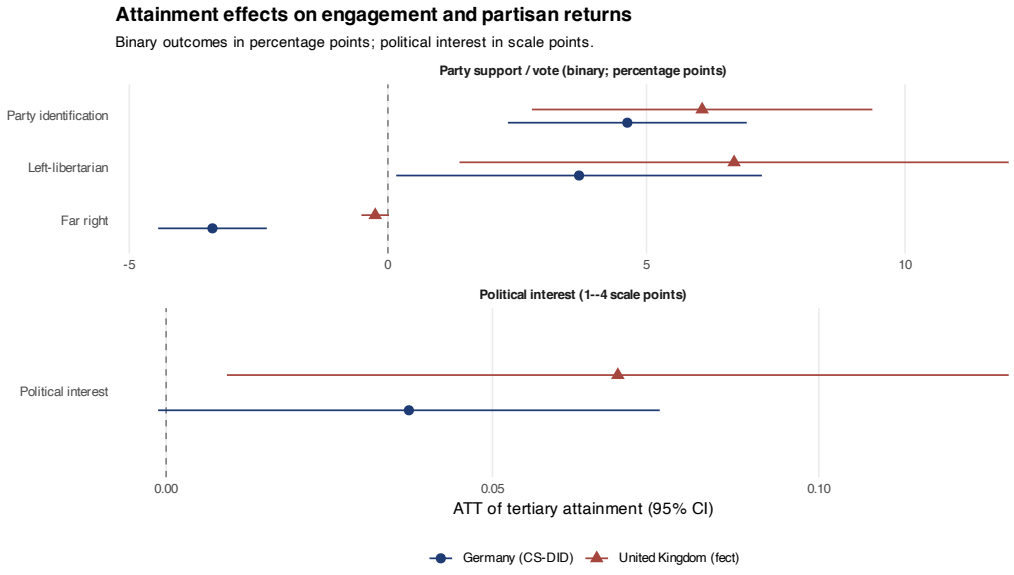


Figure 1: Attainment effects on engagement and partisan returns, Germany and the UK.

political interest is essentially constant across the propensity distribution: improvements in engagement from attainment reflect treatment effects, not selection. Graduates from backgrounds with lower or higher likelihoods of graduating make the same gains. Partisan outcomes demonstrate real change. The counterfactual effect on the untreated — the partisan shift a non-attainer would have shown had they attained — is substantially larger than the effect on the treated, and the marginal effect rises across the propensity distribution. These are two different margins and they run in opposite directions. On the ATT–TUT contrast the pattern is negative selection on unobserved gains: those who did not attain would have moved further toward the Greens than the attainers did. On observed propensity it is positive: among respondents close to the margin of indifference, those whose background already made attainment likely carry the larger return. Separating the two is what the scalar reparameterisation of Zhou and Xie (66) is for. The Brand–Xie propensity-by-treatment interaction on the same panel (Appendix Table B.4, withheld) recovers only the observed margin, which is why it reads as positive selection on its own, and it is now identified under the local-IV assumption rather than under the assumption of zero omitted-variable bias. Both margins reappear in the West-Germany subsample (Table 7, lower rows), where the unobserved-gain contrast widens: the effect on the treated is no longer distinguishable from zero while the effect on the untreated remains so. The propensity gradient is therefore not a legacy of East/West composition[4]. Full estimates for the remaining outcomes are in Appendix Table C.1, withheld, and the Bundesland fixed-effects DRDID robustness in Appendix Table C.2, withheld.

For the UK, three converging non-parametric decompositions show positive selection on political interest: graduates from advantaged backgrounds (A-level holders, early-cohort enrollees, upper propensity quintiles) improve their political interest more than the average. Negative selection appears in support for the Labour party. Among A-levels the ATT is null, yet the propensity-quintile decomposition picks up a gain among UK graduates from disadvantaged backgrounds — the same population the Labour vote-proxy ATT captures while the leaning ATT does not. UK graduates from disadvantaged backgrounds carry the support for Labour; advantaged graduates

[4]The East-only sub-panel is too small for the bootstrap intervals.

Table 7: Effects of university attainment by propensity to enter university (Zhou–Xie local-IV, Germany)

		Counterfactual averages		Marginal effect at	
Outcome	Sample	On the treated	On the untreated	Low propensity	High propensity
Political interest	All Germany	$+1.067^{***}$	$+1.053^{***}$	$+1.156^{***}$	$+0.979^{***}$
		(0.166)	(0.180)	(0.167)	(0.158)
	West only	$+0.527^{**}$	$+0.641^{***}$	$+0.668^{***}$	$+0.520^{**}$
		(0.224)	(0.231)	(0.212)	(0.210)
Leans Greens	All Germany	$+0.717^{***}$	$+0.880^{***}$	$+0.742^{***}$	$+0.930^{***}$
		(0.160)	(0.162)	(0.129)	(0.152)
	West only	$+0.205$	$+0.625^{***}$	$+0.397^{**}$	$+0.549^{***}$
		(0.202)	(0.208)	(0.155)	(0.192)

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Person-level bootstrap standard errors in parentheses (500 replicates). Zhou (65) local-IV marginal treatment effect, birth-Bundesland instrument; covariates gender, year of birth, migrant status, parental education and ISEI, HE-eligibility; propensity trimmed to $[0.02, 0.98]$.
On the treated/untreated (TT/TUT) integrate MTE over the respective propensity distributions; *low/high propensity* are MPRTE at the 25th/75th percentile. Estimates are on the local-IV latent scale.
East-only estimates and the remaining outcomes are in Appendix Table C.1, withheld.

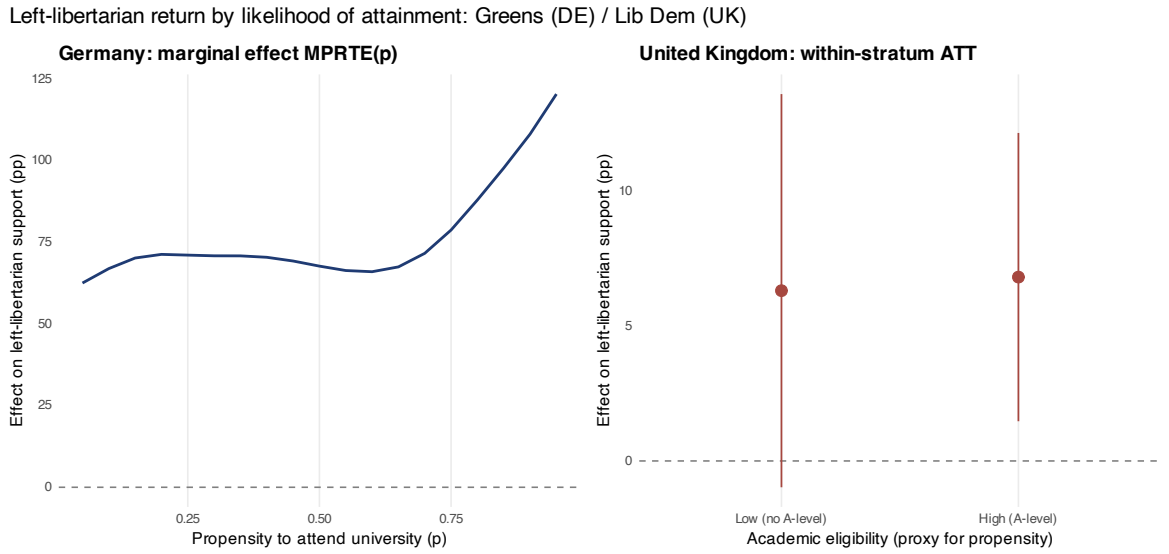


Figure 2: Heterogeneity in the attainment return by likelihood of attainment.

do not. This reproduces the negative-selection pattern from the Brand and Xie (8) HTE model for Labour (Appendix Table B.5, withheld). It is the inverse of the German Greens result on the observed-propensity margin, where the return rises with the likelihood of attaining; on the unobserved margin the two countries agree, in that both place the larger counterfactual gain among those who did not attain.

A descriptive complement checks the ATT against four moderators: gender, parental education, migrant or ethnic background, and parental occupational status (AIPW interactions). Gender is the only one that moderates the partisan return, and it does so in both countries (Figure 3). In Germany women carry the shift toward the Greens and, in the same estimates, a movement away from the SPD; in the UK they carry the shift toward Labour, while the Liberal Democrat return is flat by gender. The engagement effect is not significantly moderated by any of the four: the larger gains among migrant-background and low-parental-ISEI graduates are consistent with the resource-substitution channel of Lindgren et al. (31) but do not reach conventional significance. Read against the aggregate pattern, the moderation places the gendered movement on whichever party the electoral system makes the available socially liberal option, rather than on a particular party family.

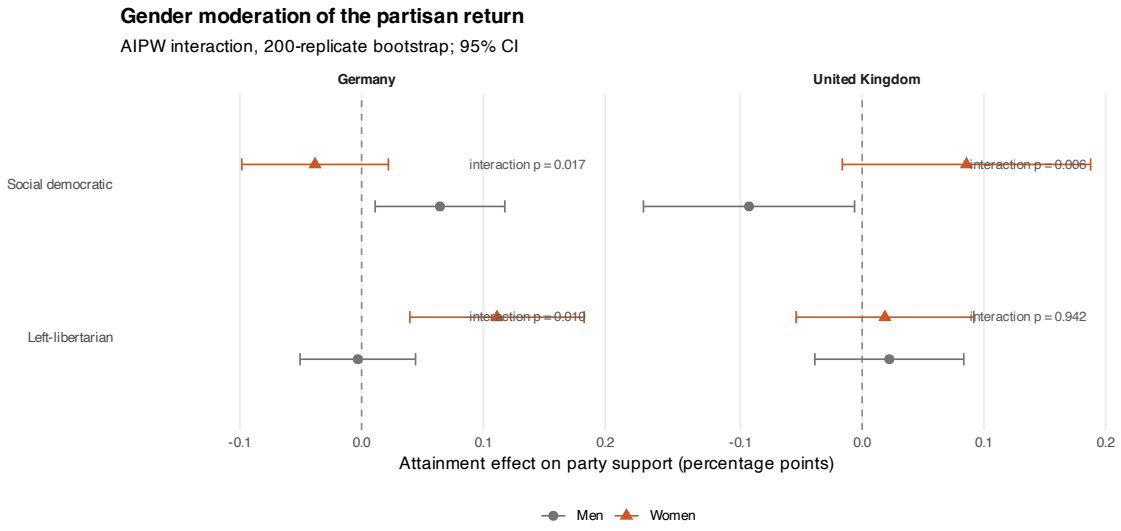


Figure 3: Gender moderation of the partisan return, Germany and the UK.

Heterogeneity by field of study (Germany)

Field of study shapes the Greens return within particular cohorts, not across the window as a whole. Pooled over all cohorts, high-CECT and low-CECT graduates shift toward the Greens by indistinguishable amounts — on the lean (Table 8) and at the ballot (Table 9) alike — but that null averages over a time-moving difference, which the cohort decomposition (Figure 4) recovers. Field content divides graduates more durably on the centre-right axis (Table 10).

Table 8: Heterogeneity by field of study (Germany, SOEP)

	Engagement		*Party support*			
Treatment	Political interest	Has party	CDU/CSU	SPD	Greens	Linke
Stratified ATT of attainment by field track (vs. non-attainers)						
Tertiary | High-CECT	$+0.070$	$+0.023$	-0.018	$+0.060$	$+0.032$	-0.066^{**}
	(0.046)	(0.031)	(0.039)	(0.041)	(0.047)	(0.029)
Tertiary | Low-CECT	$+0.100^{***}$	$+0.088^{***}$	-0.012	-0.002	$+0.047^{*}$	$+0.009$
	(0.028)	(0.018)	(0.033)	(0.029)	(0.027)	(0.014)
Between-track contrast (high-CECT − low-CECT)						
CECT contrast	-0.031	-0.065^{*}	-0.006	$+0.063$	-0.015	-0.075^{**}
	(0.054)	(0.036)	(0.051)	(0.051)	(0.054)	(0.032)

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Bootstrap standard errors in parentheses.
Callaway–Sant'Anna staggered DiD (IPW); the cohort decomposition is in Figure 4.
Stratified rows: attainment within each CECT track vs non-graduates (Panel B, Table 2). Final row: the between-track field contrast.

Table 9: Field of study and the vote-anchored partisan effect of tertiary attainment (Hooghe-extension fect, SOEP)

	Greens	SPD	CDU/CSU	Linke
High-CECT track	$+0.055$	-0.008	-0.046	$+0.016$
	(0.037)	(0.031)	(0.035)	(0.013)
Low-CECT track	$+0.031^{*}$	-0.023	$+0.026^{*}$	$+0.005$
	(0.016)	(0.015)	(0.015)	(0.013)
High − Low	$+0.025$	$+0.016$	-0.072^{*}	$+0.011$
	(0.040)	(0.035)	(0.039)	(0.018)

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Bootstrap standard errors in parentheses.
Xu et al. (62) fect on the vote-anchored composite. Treatment is tertiary attainment; controls are non-graduates within each CECT track (Panel B, Table 2). "High − Low" is the between-track field contrast. Corresponding lean estimates are in Table 8.

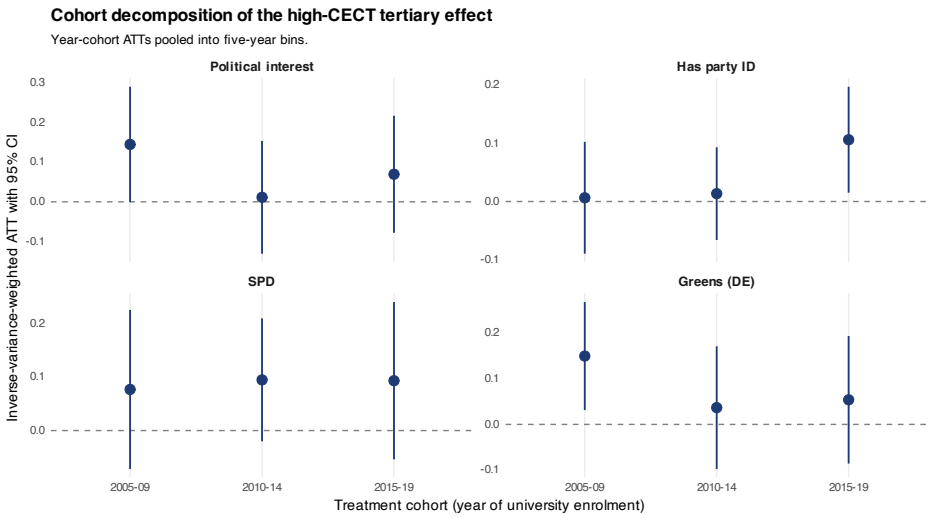


Figure 4: ATTs grouped by years of enrolment

The staggered approach (Figure 4) shows the high-CECT Green voters are concentrated in the enrolment window of 2005–09 exclusively. The first to enter after the Bologna reforms, arriving on the labour market just before climate change seriously entered the Zeitgeist. The fixed-effects estimator collapses the university experience into a single event and thus averages differences across these windows toward zero signficance. This is particularly problematic in the case of educational expansion and educational socialization effects when the outcome is also a moving target, ongoing party fragmentation, which has its own compositional effects over time (63).

The field-stratified design follows Hooghe et al. (27) on treatment definition, CECT split, and outcome construction (Appendix Table C.4, withheld). However, Hooghe et al. (27)'s replication files assign science, mathematics, and computer sciences with the communicative and cultural skill scores of teacher education, which is at odds with the technical scores the same code gives these fields on the remaining two dimensions, and with the floor-level CECT they receive in the authors' own US survey data. I return them to the technical-economic pole, and reclassify medicine there in line with its US-data scoring, so Table 8 uses the corrected split; law already sits on that side. A coarser STEM-versus-Liberal-Arts dichotomy (Table 10) gives a convergent reading: HASS graduates (humanities, arts, social sciences) shift toward the Greens at the ballot, while STEM and economics graduates show no Greens shift and instead move toward the centre-

right CDU/CSU. Neither group shifts toward die Linke or the SPD: the field effect runs along the Green/centre-right cleavage, and away from the redistributive-socialist axis. High-CECT graduates lean away from die Linke, becoming part of the culturally left yet opposed to the redistributive left. The HASS-minus-STEM contrast is clear on the centre-right axis and marginal on the Greens. Thus, the field effects run along the Green/centre-right cleavage rather than the redistributive-socialist axis.

Field content adds little to the far right; the reduction is an attainment effect. Within Germany it concentrates in the economic and technical track, whose non-attainers are the most far-right-prone — the within-country echo of the A-level gradient above — and is absent in the cultural track, whose non-attaining baseline already sits near the floor. The far right is of a piece with the Greens result: attainment moves both, while the field divergence that survives pooling is on the centre-right (Table 10).

Pre-trends, placebos, and design sensitivity

I run three tests on the staggered difference-in-differences estimates to see whether they reflect genuine effects or artifacts of the design. The first is a pre-treatment placebo. I re-estimate the same specification on political outcomes measured *before* anyone in the sample has entered higher education. If the estimator returned an apparent effect in this window — where no treatment has yet occurred — it would mean that future graduates and non-graduates were already on diverging trajectories, and that the post-treatment estimates pick up that prior divergence rather than the effect of the degree. None of the placebo estimates are significant at conventional thresholds: there are no detectable pre-trends, and the parallel-trends assumption is not violated.

The second relaxes that assumption rather than testing it, since parallel trends cannot be verified for the post-treatment period itself. I apply the bounded-violation sensitivity analysis of Rambachan and Roth (42), which asks how large a departure from parallel trends the data could tolerate before an estimated effect loses significance. The field-stratified partisan effects break down at the smallest violation — their confidence intervals already include zero — so the parti-

Table 10: Field-of-study fect estimates by content classification (SOEP, vote-or-lean composite)

	Greens	SPD	CDU/CSU	Linke	Far right
Strict HASS vs non-HASS					
HASS	$+0.097^{*}$	-0.017	-0.060	$+0.009$	-0.040^{***}
	(0.049)	(0.036)	(0.039)	(0.016)	(0.004)
non-HASS	$+0.022$	-0.024	$+0.027$	$+0.009$	-0.015^{*}
	(0.017)	(0.015)	(0.017)	(0.012)	(0.006)
HASS − non-HASS	$+0.074$	$+0.008$	-0.088^{*}	$+0.000$	-0.025^{***}
	(0.052)	(0.039)	(0.043)	(0.020)	(0.007)
HSS vs STEM + Economics					
HSS	$+0.083^{\dagger}$	-0.015	-0.043	$+0.007$	-0.036^{***}
	(0.043)	(0.031)	(0.033)	(0.013)	(0.005)
STEM + Economics	$+0.023$	-0.026	$+0.027$	$+0.010$	-0.017^{**}
	(0.017)	(0.016)	(0.017)	(0.011)	(0.006)
HSS − STEM	$+0.061$	$+0.010$	$-0.070^{\dagger}$	-0.003	-0.019^{*}
	(0.046)	(0.035)	(0.037)	(0.017)	(0.008)

$^{\dagger}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Bootstrap standard errors in parentheses (200 replicates).

Xu et al. (62) fect on the vote-or-lean composite. Treatment is tertiary attainment; controls are all non-graduates (Panel A′, Table 2), with the treated split by field content. Each field row is the attainment ATT for that group; the difference row is the between-field contrast (Panel C).

Classification (55): HASS = humanities, arts, social studies (edutypes 2, 3, 10); HSS adds teacher training (6); STEM + Economics = technical, agriculture, science/maths/computing, medicine, economics, law, transport (4, 5, 7, 8, 9, 11, 14).

Far right is a placebo: both sides reduce it, consistent with an attainment rather than a content effect.

Table 11: Hypotheses and verdicts

	Verdict
H1	Supported. Attainment effects are clear for party support and modest for engagement.
H2	Supported at the ballot. Graduates' votes move to the Greens (Germany) and Liberal Democrats (UK), not the redistributive centre-left or die Linke, and away from the far right.
H3	Supported. Political returns are heterogeneous in party support — positive selection on the Greens in Germany, negative selection on Labour in the UK.
H4	Supported in a specific form. The field difference on the Greens is time-bounded — high-CECT graduates carry the shift in the 2005–09 cohort, averaged to null when pooled — while the divergence that survives pooling is on the centre-right: HASS graduates move away from the CDU/CSU, STEM and economics graduates toward it.

san field claim does not rest on these difference-in-differences intervals but on the vote-anchored within-completer design and the cohort decomposition, neither of which invokes parallel trends. Engagement is more robust, and the aggregate attainment effects tolerate moderate violations. A reader who doubts parallel trends to that tolerance can rely instead on the marginal-treatment-effect results (section), identified through the birth-region instrument rather than parallel trends.

The third test randomizes treatment timing. I reassign the timing of the educational transition among those eligible for it, producing samples in which any estimated effect is spurious by construction, and re-estimate fifty times. A valid test at the 5% level should reject this fabricated null about 5% of the time. For four of the six outcomes the placebo rejection rate stays near nominal; for the Greens and the centre-left it runs higher, meaning the conventional standard errors for those two outcomes are somewhat too small. Once corrected, the all-Germany Greens effect still clears significance, while the West-Germany treatment-on-the-treated estimate becomes borderline.

Discussion

The political returns to higher education are mediated by the electoral system within which it operates: university graduates flow into socially liberal alternatives where the electoral system makes them viable, and into the dominant centre-left where it does not. The cross-country aggregates (Table 3) demonstrate that there are causal movements towards the Greens in Germany and the Liberal Democrats in the UK, with no significant effects for Labour or the SPD. However, only in Germany does this support translate into a vote, whereas Labour instead pick up the Liberal Democratic supporters. The pattern is not about how different educational systems produce different attainment effects; it is about how similar graduate dispositions are filtered by the party system into opposing logics of voting behaviour. What this cross-country contrast obscures in its aggregation of effects, is who within the graduate population carries each effect. Disaggregating by likelihood of attainment in § sharpens the picture: the same electoral-system mediation operates on different graduate populations in the two countries.

The cross-country pattern refines rather than restates the Brahmin-left account of the educational cleavage: the educational divide does not operate uniformly through green and left-libertarian parties, but through whichever socially liberal alternative the electoral system makes electorally viable. In Germany we saw a positive selection towards greens, the wealthy and highly educated were more likely to support the Greens based on their priors and eventually do so, while the SPD attracts party support but not actual votes. In the UK negative selection appears for Labour supporters, whose support is driven by disadvantaged graduates, with a small positive selection slope for the Liberal Democrats. The Greens and the Liberal Democrats attract similar voters, predominantly privileged graduates, but only one of these parties enjoys the confidence of its supporters to secure seats. In other words, the Brahmin vote manifests where the electoral system allows it (1). Moreover, the persistence of the highly educated social-democratic voter is the social reproduction of the petite bourgeoisie (4): advantaged graduates reproduce their parents' socially liberal disposition within higher education and translate it into a culturally liberal, economically conservative vote, in contrast to the United Kingdom, where disadvantaged graduates

reproduce gains for the Labour party. It appears that in terms of political behaviour, the educational divide does not operate through class reproduction alone. Where a socially liberal alternative is electorally viable, the cultural dispositions become votes; where they are not, this support turns to the centre-left. Both the class-reproduction account and the electoral-viability mediation treat the credential as a single object whose return is then filtered externally. The field-of-study results in § show how a credential does not produce uniform effects. The content of the degree carries its own partisan effect.

Restricting to those that do obtain a credential, field of study shapes the Greens return within particular cohorts rather than across the window as a whole. Pooled over all cohorts, the within-completers contrast on the Greens is null — in leaning (Table 8) and at the ballot (Table 9) alike. But that null averages over a time-moving difference: the cohort decomposition (Figure 4) locates the high-CECT Greens shift in the 2005–09 enrollment cohort, which arrived on the German labour market as climate and sustainability began taking up a larger position in the Zeitgeist, with the low-CECT contribution arriving later. Aggregate estimators such as the approach used in Hooghe et al. (27) pool attainment effects across enrollment cohorts and across tracks, so a period-concentrated, field-specific effect is averaged into an apparent null rather than recovered — a mismatch between the interval an estimator imposes and the interval over which an effect operates, not its absence. Where field content does divide graduates is the centre-right (Liberal Arts versus STEM, Table 10): STEM and economics graduates move toward the CDU/CSU and Liberal Arts graduates — humanities, arts, social sciences, and teacher training — away from it, while neither group moves toward the Linke or the SPD. The credential carries the broad partisan return; the content of the field shapes which cohorts carry it and sorts graduates along the established left–right divide.

The humanities-and-social-sciences shift and the STEM null are a substantive contrast the design identifies but does not mechanistically adjudicate. Two candidate channels each have independent literature support: a curricular-content account that humanities and social sciences cultivate contextual and civic reasoning the natural sciences do not (21; 24), and an occupational-

sorting account that Liberal Arts and STEM graduates enter different labour-market segments — the sociocultural professional fraction in one case and private-sector technical roles in the other — with different work logics (37). The present design tests neither channel directly, and selection on pre-treatment dispositions remains a third candidate that the within-completers comparison constrains without eliminating. How either the credential or this curricular contrast distributes across the structural conditions of access to higher education is what the propensity decomposition in § reveals.

Heterogeneity in the attainment effect appears in partisan direction, not in engagement. Of the moderators that also predict attainment: gender, parental education, migration or ethnic background, and parental occupational status, only gender distinguishes the partisan shift, which is carried by women. The engagement gains, by contrast, are not significantly moderated by any of these characteristics: the larger gains among migrant-background and lower-SES graduates point toward the resource-substitution logic of Lindgren et al. (31), whereby the university supplies the political articulation a constrained adolescence did not, but they do not reach conventional significance, and the local-IV estimates likewise leave the engagement return essentially flat across the propensity distribution. The university engagement is shared across demographics, while the increased support towards left-libertarian parties is gendered.

The credential and the curriculum operate on different political objects. The credential produces a modest engagement gain in both countries and a leftward lean toward the socially liberal wing of the party system; this directional effect is uniformly distributed across structural moderators, selection patterns on propensity run in opposite directions for the two countries' relevant parties, and the field-stratified comparison within tertiary is null on the size of the lean. The curriculum, in turn, divides graduates along the centre-right, with communicative-cultural fields moving away from the CDU/CSU, and the technical and economic fields toward it. This relfects a division the attainment alone does not produce. Brady et al. (6) theorised that political participation is governed through increased resources, demonstrating that civic skills and peers drive participation independently of political interest. This paper arrives at similar conclusion from a different

perspective. The political returns are carried by attainment, yet this effect is time bounded towards the socially liberal wing, while the content of the field redraws the established left–right divide rather than producing the return on its own. Future work on education and democracy should disaggregate the credential from the curriculum, account heterogeneity in the access to education, and resist treating civic improvements as onesided, when the largest political returns are conditional on *what is taught and to whom*.

Three limits define the present design. The first concerns parallel trends: the bounded-violation sensitivity of Rambachan and Roth (42) holds at the five-year post horizon under a breakdown value of $\bar{M} = 0.5$; a reader unwilling to assume parallel trends to that tolerance should treat the staggered DiD estimates as suggestive and rely on the local-IV evidence for the German heterogeneity results. The second concerns the random-reassignment placebo, which shows mild over-rejection on the Greens and centre-left outcomes; the all-Germany Greens effect survives size correction, but the within-West local-IV estimate becomes borderline and I treat it as such. The third concerns the instrument itself. The region-at-birth (Bundesland) instrument works for Germany; the same cannot be done for the UK given data constraints, and the UK heterogeneity layer rests instead on the non-parametric three-way decomposition. A compulsory-schooling instrument in a comparable national context within the same period is the natural extension of this paper.

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Appendix A. Estimators and identifying assumptions

This section provides the formal specifications of the estimators used in the article; see section .

A.1 Staggered difference-in-differences

The approach follows Callaway et al. (9) and is used for the SOEP aggregate effect on engagement and party-support outcomes (section (a)) and for the field-of-study heterogeneity (section (d)).

Let $G_i \in \{2000, \ldots, 2020\} \cup \{0\}$ denote the calendar year of respondent i's first reported tertiary enrolment, with $G_i = 0$ for never-treated respondents. The group-time average treatment on the treated is

$$\mathrm{ATT}(g,t) = E\big[Y_{it}(1) - Y_{it}(0) \,\big|\, G_i = g\big], \quad t \geq g. \tag{1}$$

Under conditional parallel trends and no-anticipation (9, Assumptions 3–4), $\mathrm{ATT}(g,t)$ is identified by the inverse-probability-weighted estimator

$$\widehat{\mathrm{ATT}}(g,t) = E\left[\left(\frac{1\{G_i = g\}}{P(G_i = g)} - \frac{p_g(X_i)\,(1 - D_i^g)}{\big(1 - p_g(X_i)\big)\,P(G_i = g)}\right)\big(Y_{it} - Y_{i,g-1}\big)\right], \tag{2}$$

where $p_g(X) = P(G_i = g \mid X_i, G_i \in \{g, 0\})$ is the generalised propensity score, X_i contains gender, parental education, migrant background, and year of birth, and $D_i^g = 1\{G_i = g\}$. The overall ATT is the simple aggregation across groups and post-treatment periods,

$$\mathrm{ATT}_{\text{overall}} = \frac{1}{|\mathcal{T}|} \sum_{(g,t)\in\mathcal{T}} \mathrm{ATT}(g,t), \tag{3}$$

with $\mathcal{T} = \{(g,t) : t \geq g\}$. Inference uses a 999-replicate multiplier bootstrap. Identification rests on conditional parallel trends between treated cohort g and the never-treated comparison group at every post-treatment t, no anticipation in pre-treatment periods, and the standard overlap condition $0 < p_g(X) < 1$.

The no-anticipation assumption is testable through the pre-treatment event-study coeffi-

cients $\widehat{\mathrm{ATT}}(\tau)$ at $\tau < 0$, which should be jointly indistinguishable from zero. In the present application the pre-trends are flat for partisan-direction outcomes and the vote-or-lean composite, but show significant pre-treatment ATTs for engagement outcomes. HonestDiD sensitivity (42) is reported on every outcome to quantify the violation of parallel trends under which the post-treatment ATT remains distinguishable from zero.

A.2 Interactive fixed effects

The approach follows Xu et al. (62) and is used for the UK aggregate effect on all outcomes (section (a)) and the vote-or-support behavioural extension in both panels (section (e)).

The untreated outcome is modelled as

$$Y_{it}(0) = \alpha_i + \xi_t + \lambda_i^\top f_t + X_{it}^\top \beta + \varepsilon_{it}, \tag{4}$$

where α_i is a person fixed effect, ξ_t a wave fixed effect, $f_t \in R^r$ a vector of latent time-varying factors, $\lambda_i \in R^r$ the person-specific factor loadings, X_{it} contains gender, ethnicity (UK) or migrant background (Germany), parental education, and year of birth, and ε_{it} is the error term. Factor rank r is selected by cross-validation; $r = 0$ recovers two-way fixed effects without latent-factor structure. The counterfactual $\widehat{Y}_{it}(0)$ is imputed under nuclear-norm regularisation. Treatment D_{it} is absorbing: $D_{it} = 1$ from the first wave a respondent is observed in the treated state. The counterfactual for treated observations is imputed by fitting (4) on the never-treated subsample and predicting the treated sample under the same parameters. The ATT, with $\mathcal{T}_1$ the set of treated observations with at least one pre-treatment observation in the UK and at least five in Germany, is

$$\widehat{\mathrm{ATT}} = \frac{1}{|\mathcal{T}_1|} \sum_{(i,t)\in\mathcal{T}_1} \big(Y_{it} - \widehat{Y}_{it}(0)\big). \tag{5}$$

Standard errors are obtained from a 200-replicate bootstrap. The covariate vector X_{it} enters the outcome model: covariates adjust for observed confounders in the counterfactual prediction, while

the latent factor term $\lambda_i^\top f_t$ absorbs unobserved time-varying confounders. This is consequential because earlier work has found that unobserved time-varying confounders can account for the bulk of the estimated effect (e.g. 65).

Identification requires: (i) individual and time fixed effects absorb all time-invariant variation and common shocks; (ii) the latent factor absorbs unobserved time-varying confounders correlated with treatment; and (iii) the counterfactual fitted on never-treated respondents extrapolates correctly to treated respondents absent treatment. The third assumption is tested in a placebo equivalence test that holds the cross-validated factor rank fixed and reserves event-times $\tau \in \{-2, -1\}$ out of the ATT computation; the parallel-trends assumption is supported if the placebo ATT on the reserved periods is statistically indistinguishable from zero ($p > 0.10$).

A.3 Marginal treatment effect

The approach follows Zhou and Xie (66) and is used for the Germany heterogeneity-by-likelihood layer (section (b)).

Treatment selection is modelled through a latent-index threshold-crossing rule (22). Each respondent is characterised by two quantities. The propensity score,

$$P(Z_i, X_i) = P(D_i = 1 \mid X_i, Z_i)\,, \tag{6}$$

is a scalar summary of observed characteristics X_i (gender, year of birth, migrant background, parental education, parental occupational status, Abitur eligibility) and the local instrument Z_i (Bundesland of birth, categorical). The latent resistance quantile is

$$U_i = F_{V|X}(V_i) \sim \text{Uniform}(0, 1), \tag{7}$$

where V_i is unobserved individual resistance to treatment. The selection rule is then

$$D_i = 1\{P(Z_i, X_i) > U_i\} : \tag{8}$$

a respondent is treated if and only if observed propensity exceeds unobserved resistance.

The original MTE of Heckman and Vytlacil (22), $\mathrm{MTE}(x,u) = E[Y_i(1) - Y_i(0) \mid X_i = x, U_i = u]$, conditions on the full covariate vector and is high-dimensional. Zhou and Xie (66) reparameterise by replacing X_i with its scalar propensity-score summary,

$$\widetilde{\mathrm{MTE}}(p,u) = E\big[Y_i(1) - Y_i(0) \,\big|\, P(Z_i, X_i) = p,\ U_i = u\big]\,. \tag{9}$$

Because (P, U) jointly determines treatment status, $\widetilde{\mathrm{MTE}}$ captures all of the heterogeneity that contributes to selection bias.

Three estimands emerge from weighted integrals of $\widetilde{\mathrm{MTE}}$ over the joint distribution of (P, U). The ATE under random assignment is

$$\mathrm{ATE} = \int_0^1 \int_0^1 \widetilde{\mathrm{MTE}}(p,u)\, dF_{P,U}. \tag{10}$$

The ATT, the expected effect among respondents who actually received treatment ($u < p$), is

$$\mathrm{ATT} = \int_0^1 \int_0^p \widetilde{\mathrm{MTE}}(p,u)\, dF_{P,U|D=1}. \tag{11}$$

The treatment effect on the untreated (TUT), the counterfactual effect among those with $u \geq p$, is

$$\mathrm{TUT} = \int_0^1 \int_p^1 \widetilde{\mathrm{MTE}}(p,u)\, dF_{P,U|D=0}. \tag{12}$$

The contrast $\mathrm{ATT} - \mathrm{TUT}$ has a direct interpretation: $\mathrm{ATT} > \mathrm{TUT}$ indicates positive selection on gains, while $\mathrm{ATT} < \mathrm{TUT}$ indicates negative selection on gains (8).

The marginal policy-relevant treatment effect at propensity level p is

$$\mathrm{MPRTE}(p) = \widetilde{\mathrm{MTE}}(p,p), \tag{13}$$

the expected effect among respondents at the margin of indifference, those an incremental policy

change at level p would move in or out of treatment. $\mathrm{MPRTE}(p)$ decomposes into an observed-heterogeneity component (variation in p at fixed u) and an unobserved-gain component (variation in u at fixed p), and the two need not agree in their respective sign. Selection on unobserved gains is read off the ATT–TUT contrast above: $\mathrm{ATT} < \mathrm{TUT}$ is negative selection, a situation where those that were not treated could have gained more than the treated did. Selection on observed propensity is read off the slope of $\mathrm{MPRTE}(p)$: rising in p places the larger marginal return among respondents whose observed characteristics already made attainment likely. A design can exhibit both at once, and the German Greens estimates do. Reporting only one of the two margins is what the scalar propensity-score reparameterisation of Zhou and Xie (66) is meant to prevent.

Identification rests on: (i) the exclusion restriction, Z_i affects political outcomes only through D_i conditional on X_i; (ii) instrument relevance, Z_i shifts the propensity score conditional on X_i; and (iii) additive separability of $Y_i(d)$ in X_i and U_i, plus standard local-IV support conditions on the joint density of (P, U). The exclusion restriction is plausible in the German sample because Bundesland of birth is time-invariant and affects regional university access and exposure to academic trajectories; the relevance condition is verified by a likelihood-ratio test on the sixteen Bundesland indicators ($\chi^2_{15} = 116.22$, $p < 10^{-6}$). The parallel UK pipeline using a tuition-fee-shock instrument fails the relevance test ($\chi^2_3 = 2.62$, $p = 0.45$), consistent with the price-inelasticity of UK enrolment documented in Murphy et al. (35); the stratification-based heterogeneity estimation (section (b)) is used for the UK instead.

A.4 Augmented inverse-probability-weighted interaction

The augmented inverse-probability-weighted (AIPW) interaction estimator is used for the heterogeneity-by-structural-predictors layer in both panels (section (c)). For each moderator $M \in \mathcal{M}$ and outcome Y, the within-stratum ATT is estimated separately on the strata $M = 0$ and $M = 1$:

$$\widehat{\mathrm{ATT}}_{M=m} = E\left[\mu_1(X) - \mu_0(X) + \frac{D - e(X)}{e(X)\big(1 - e(X)\big)}\big(\Delta Y - \mu_D(X)\big) \,\Big|\, M = m\right], \tag{14}$$

where $\mu_d(X)$ is the outcome regression of within-person change $\Delta Y = Y_{\text{post}} - Y_{\text{pre}}$ on X in stratum $\{D = d\}$, $\mu_D(X) = D \cdot \mu_1(X) + (1 - D) \cdot \mu_0(X)$, and $e(X) = P(D = 1 \mid X)$ is the propensity score. The moderation contrast is

$$\Delta_M = \widehat{\mathrm{ATT}}_{M=1} - \widehat{\mathrm{ATT}}_{M=0}, \tag{15}$$

with standard errors from a 200-replicate non-parametric bootstrap. The estimator is doubly robust (44): consistent for $\mathrm{ATT}_{M=m}$ under correct specification of either μ_d or $e(X)$. The moderator set is gender, high parental education (mother and father separately), high parental occupational status, and ethnicity/migrant background.

Appendices B and C

This confernece circulation only contains the first Appendix, A. The remaining appendices: the sample and propensity-score diagnostics, the parallel-trends and pre/post window sensitivity tables, the full marginal-treatment-effect grid by East/West subsample, the instrument-validity check, the CS-DiD robustness specifications, and the replication of Hooghe et al. (27) are available from the author on request and accompany the submitted version. Cross-references in the text to tables in those appendices are marked *withheld*.